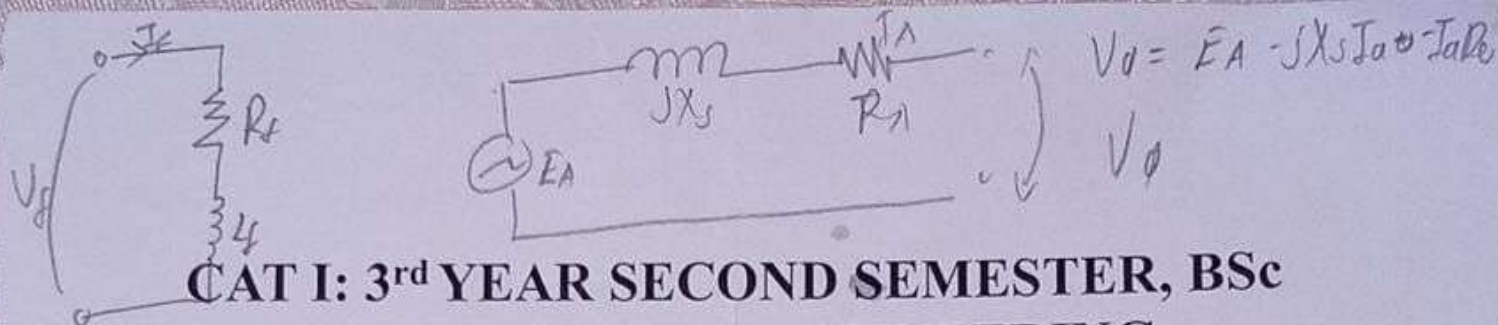


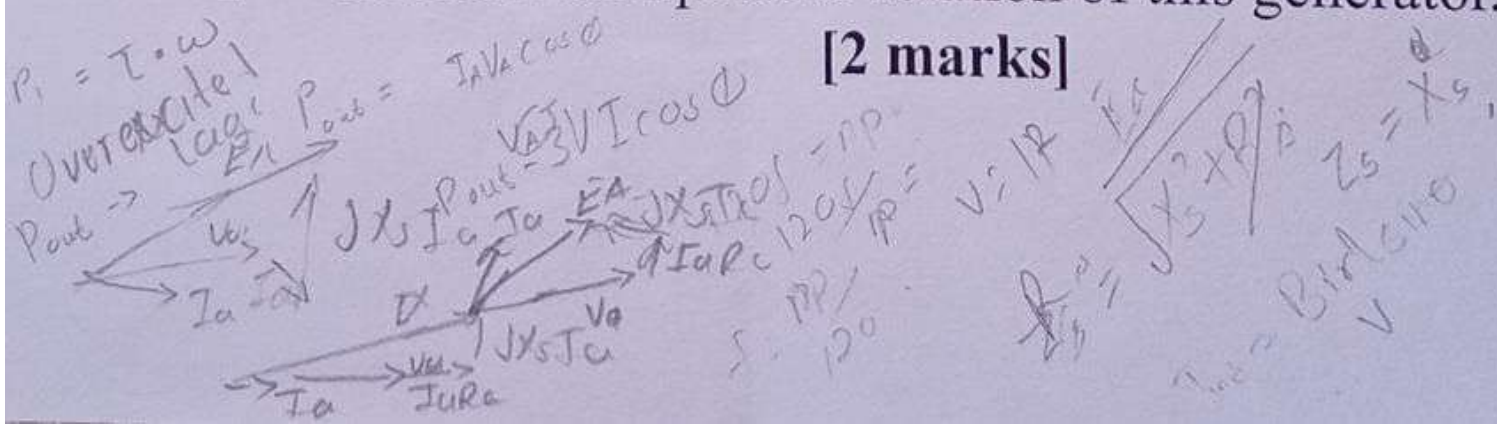


**CAT I: 3rd YEAR SECOND SEMESTER, BSc
MECHATRONIC ENGINEERING
EMT 3204: ELECTRICAL MACHINE DRIVES II**

- A. With the aid of sketches, discuss the rotating magnetic field concept of a stator containing three coils, each 120° apart. **[12 marks]**
- B. Name the four factors that affects the torque generated by real ac machines **[4 marks]**
- Strength of the rotor mag
- Strength of external "
- C. State and explain the four conditions to be met before paralleling synchronous generators. **[4 marks]**
- D. A 480-V, 50-Hz, Y-connected, six-pole synchronous generator has a per-phase synchronous reactance of 1.0 ohms. Its full-load armature current is 60 A at 0.8 PF lagging. This generator has friction and windage losses of 1.5 kW and core losses of 1.0 kW at 60 Hz at full load. Since the armature resistance is being ignored, assume that the I^2R losses are negligible. The field current has been adjusted so that the terminal voltage is 480 V at no load.

I. Calculate the speed of rotation of this generator.

[2 marks]



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$$V_d = E_a - I_a X_s - R_a I_a$$

II. Find the terminal voltage of this generator if the following are true —

▪ It is loaded with the rated current at 0.8 PF lagging. **[1 mark]**

▪ It is loaded with the rated current at 1.0 PF. **[1 mark]**

▪ It is loaded with the rated current at 0.8 PF leading. **[1 mark]**

III. Find the efficiency of this generator (ignoring the unknown electrical losses), when it is operating at the rated current and 0.8 PF lagging. **[2 marks]**

IV. Find the voltage regulation of this generator at :

• 0.8 PF lagging **[1 mark]**

• Unity PF. **[1 mark]**

• 0.8 PF leading. **[1 mark]**

$$VR = \frac{V_{nl} - V_{fl}}{V_{fl}} \times 100\%$$

VR